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Tackling Well Drifting Issues When Drilling Deep Gas Wells

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Abstract

When drilling a deep vertical well, there is always a small inclination build-up that occurs naturally due to dip angles of formations, stiffness of Bottom Hole Assemblies (BHAs), and applied drilling parameters. Sometimes dip angles can be extreme, which can lead to unwanted inclination build-up or drifting, due to which, the well-bore lands significantly away from the intended target location. When planning a well, it is necessary to know beforehand whether drifting will be an issue or not.

A thorough study of wells drilled in various fields was conducted using available digital technologies, which assisted in compilation, synthesis, and interactive graphical evaluation to aid in the identification of drift-prone areas. In addition to the available data on formation dip angles, this approach presented an effective alternative to avoid costly correction runs later. During this analysis, the types of BHAs used in a particular interval of the wellbore were also studied, and the most effective BHA type was identified. Appropriate drilling strategies were outlined to develop a road-map for drilling in drifting-prone areas.

Based on the results of this study, it was identified that in surface holes, a short-packed BHA would prevent any inclination build-up, although a semi-packed BHA is prone to inclination build-up due to less rigidity. In intermediate and deeper sections, it was observed that pendulum BHAs are the best choice to maintain verticality as they tend to counter any drifting tendency. In many instances when packed BHAs were deployed to maintain angle, it was observed that inadvertent drifting still occurred even with controlled drilling parameters, which is counter-intuitive to traditional drilling practices. This approach was then applied to drilling a deep gas well with good success. It helped reduce the costs associated with wellbore deviation surveying tools, which were previously typically run in shallow sections to monitor deviation. In this well, BHAs were run as per the heat map recommendations, and the well remained vertical all the way to final Total Depth (TD) as confirmed by gyro and Measurement-while-Drilling (MWD) measurements later in the drilling phase. This established the validity of the heat map model, which can be used in future wells with confidence and will lead to time and cost savings.

The use of digital technologies enables efficient data analysis, from which key insights into drilling operations can be obtained. These help drilling engineers in the design phase to plan drilling BHAs in the planning phase and avoid having to deal with inadvertent drifting issues during well construction operations. In this way, such technologies are enablers for operational efficiency, bringing significant time and cost benefits.

Background and Introduction

Drilling gas wells sometimes entails a variety of challenges. The targeted reservoirs may be characterized by high-pressure, high- temperature (HPHT) conditions, presence of natural fractures and loss zones, unstable shale formations, complex lithologies, and significant depths that exceed conventional drilling limits. Drilling into these formations often requires navigating intricate geological features such as salt domes, fractured carbonate zones, and tight formations with variable porosity, permeability, and dip angles. Many efforts are made, and innovative solutions are deployed to overcome challenges in deep gas drilling; however, some go unnoticed. From rotary steerable systems (RSS) to advanced mud systems and real-time monitoring, the Oil and Gas Industry has continually pushed the boundaries of drilling performance. However, wellbore stability and trajectory control remain critical engineering concerns.

Wellbore Drift

Wellbore drift refers to the unintended deviation of the drilling trajectory from its planned course. This deviation can occur due to geological and operational factors, and if not addressed, it can lead to reduced reservoir contact and increased costs due to corrective actions that may see the well being plugged back and re-drilled with adequate directional control. The underlying cause of wellbore drift can be attributed to geological factors, particularly the dip angles and presence of heterogeneous formations prevalent in deep gas reservoirs. Combined with an improper BHA design and inefficient drilling parameters applied during the drilling process (operational factors), drifting can occur without being noticed in the absence of inclination measuring devices.

Wellbore drift most commonly occurs in vertical wells, where it may go undetected if survey tools are not incorporated in the BHA. In these scenarios, the walk-away from vertical of the well comes to light only when a gyro or other survey is conducted post-drilling, at which point corrective actions may be planned to correct the path. If directional surveys are available during drilling, as may be the case when drilling in pad wells or in close proximity to offset wells which necessitates monitoring real time surveys due to anti-collision issues, then drilling parameters may be adjusted to mitigate drifting tendency. This entails reducing the applied weight on bit (WOB) and string rotation per minute (RPM). Reducing WOB will cause less bending of the BHA, thereby improving the pendulum effect and allowing gravity to pull down the drilling assembly, allowing it to come back to vertical. Reducing RPM will ease the stiffening effect of the BHA again, allowing the gravity effect to take over. It is worth mentioning here that conventional wisdom dictates the use of stiff or packed BHAs containing 3 or more stabilizers to maintain verticality while drilling; however, this has been shown to be counter- productive in specific formations in the gas fields. Stiff BHAs have been observed to build inclination, which leaves many question marks as to the dynamics of drilling and the dip angles of the formations. Running formation evaluation logs in these non-reservoir sections has the potential to yield valuable information and aid Drilling Engineers in the planning phase; however, their scope is restricted due to economic reasons. On the contrary, utilizing pendulum BHAs has been shown to maintain verticality, which will be discussed in more detail later.

In case real-time surveys are not available during drilling and the wellbore drift is only identified after the section has been drilled, then the corrective actions to fix the issue include utilizing a motor with bend or RSS to correct the well path in the subsequent section if sufficient footage is available and if anticipated doglegs permit. As long as the well is drilled within the available window of target tolerance, then the production will not be affected. In worst-case scenario where the Geological objective may get compromised based on the projected placement of the wellbore in the reservoir, it may become necessary to plug back the section from a shallow depth and re-drill with adequate directional control to maintain verticality.

Although drifting does not occur in deviated wells because of the presence of directional tools and availability of regular directional surveys, which prompt correction as soon as a divergence from plan is observed, high dip angles or bedding planes can cause directional tools to give an improper response to the imposed drilling parameters and settings. It has been observed that sometimes rotary steerable systems struggle to build, drop, or turn a well path if the formation tendency is aggressive and works against the directional tools' capabilities, even if the steering ratio has been set at 100%. In such scenarios, drilling has to revert to more conventional means of directional work, such as utilizing a mud motor with a high bend setting, which can provide a better response in sliding mode. This paper addresses wellbore drifting issues in vertical deep gas wells only.

As indicated earlier, the consequences of drifting can vary in their intensity. Depending upon the available target tolerance on top of the reservoir, it may be decided to continue drilling without any remediation if sufficient margin is available, or conversely, the well may have to be plugged back and re-drilled. The following are enumerated as possible consequences of wellbore drifting:

- Extended non-productive time (NPT) due to corrective measures such as plug back and re-drill.
- Reduced rate of penetration (ROP) while controlling drilling parameters to drop inclination, impacting project timelines and cost
- Higher operational costs due to additional tools and services required to rectify drift.

Rotary steerable systems configured to maintain verticality may be utilized in extreme cases, although they come at a significant extra cost. However, their use may be justified in certain extreme cases.

Software Application Development for Tackling Wellbore Drifting

A good BHA design is fundamental to countering the drifting problem. As previously stated, it is common industry understanding that a packed or stiff BHA must be deployed to avoid any undesired or uncontrolled increase in inclination angle. This is because such a BHA has several contact points with the wellbore wall that prevent the drilling assembly from flexing, although the added risk is that of a pack-off and stuck pipe. However, counter-intuitively, certain formations display the reverse effect. It has been observed that a pendulum assembly maintains verticality better than a packed BHA. In the absence of log or seismic data to study the formation dip angles, it can only be deduced empirically that the formation tendency and dip angles are such that the flexibility offered by a pendulum BHA reacts in a productive manner and leads to near-vertical wellbores. This has been observed in several wells drilled in the gas fields, a more detailed analysis of which is provided below.

To gain better insights into this phenomenon, Drilling Engineers looked into the deviation surveys of vertically planned hole sections in gas wells, to identify which of these encountered drifting. In addition, drilling BHA data, composed of each component in the BHA, was retrieved for those hole sections. This facilitates the performance of further analysis by BHA types, and the exclusion of any BHA runs that utilized directional tools such as RSS or mud motor. As these hole sections were intended to be vertical, the maximum angle observed in their deviation surveys serves as a good proxy for drift tendency. Wellbores with angles exceeding a threshold of 12 degrees were eliminated to reduce the likelihood of single data points having a huge influence on the model, and to eliminate any wells where the BHA data was incorrectly entered, as it is rare that a hole section is planned with such an angle.

Given the availability of a large number of wells, a statistical approach was deemed suitable to analyze this phenomenon. Due to the extensive database available, a need was felt to apply advanced data analysis techniques, harnessing the power of available digital solutions to quickly sort data and analyze trends. A manual approach in this regard would be cumbersome, extremely time-consuming, and yield only limited results.

While performing statistical analysis, the aim was to find a suitable model that captures localized patterns among the wells, and at the same time, is not prone to overfitting to random patterns. With the data being composed of x and y UTM coordinates of each well, and its corresponding maximum angle, some of the advanced machine learning models, such as Random Forest, overfit vertical and horizontal patterns.

Moreover, transformations done on the variables to reduce these patterns increase the complexity of the model and reduce its interpretability. Other simpler non-linear models suffered from other issues. For example, regression splines exhibited peculiar boundary behavior. As a result, it was decided that a distance-weighted K-nearest Neighbor (KNN) model was suitable for this application, as it effectively captures localized patterns and is due to its good interpretability. Distance weighted KNN estimates the drift tendency at each grid point using the following formula:

$$Drift\,Tendency = \frac{\sum_{i=1}^{k} w_i * \theta_i}{\sum_{i=1}^{k} w_i}$$

Where

$w_i = \frac{1}{d_i}$

θ= maximum angle

d_i: Distance between grid and well

By varying the number of neighbors in the model and using cross-validation, it was determined that the highest accuracy is achieved with 3 neighbors. It was observed that this trend is localized and does not follow any identifiable pattern. Certain areas of the Gas Field are more affected by drifting than others. Same is the case in other fields. Figure 1 below is a plot of an area excluding other fields.

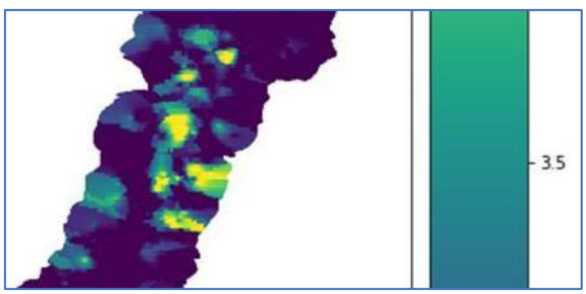


Figure 1—Drifting Tendency Heat Map of the area

Using the output of the KNN model, the drift tendency of each grid point in the field was visualized through the heatmap, enabling drilling engineers to demarcate zones based on their risk for drifting. This state-of-the-art emerging technology enables drilling engineers to make quick decisions based on the risk ranking of wells and select the appropriate BHAs. Figure 2 below shows such a heatmap with color codes for the severity of drifting risk, which was generated for this area.

Figure 2—Map of the area

To identify any relationships between drift tendency among the wells and the BHA types used while drilling, the types of BHA deployed were classified as being either a slick assembly used for clean-out purposes only, a pendulum assembly based on the distance between the bit and the first stabilizer, a packed assembly with three or more stabilizers, or a directional assembly consisting of RSS or mud motor. The pendulum and packed BHAs may further be classified as short pendulum or short packed BHA based on the length of the first drill collar. The use of the term ‘short’ refers to the presence of a pony drill collar (15 ft) being the first drill collar in the string. See Appendix 1 for illustrations of these BHA types. As mentioned earlier, hole sections drilled with a directional assembly were excluded from the analysis.

A KNN model was fit for each of these types, and their outputs were plotted in heatmaps as well, to visualize any BHA-specific drift tendency patterns. The large number of runs available for analysis enabled a like-for-like comparison of the different BHA types. BHA- specific drift tendency heatmaps are shown in Figure 3 – 5 below. This highlights the previously stated phenomena of packed BHAs suffering from drifting while short pendulum BHAs being able to maintain verticality.

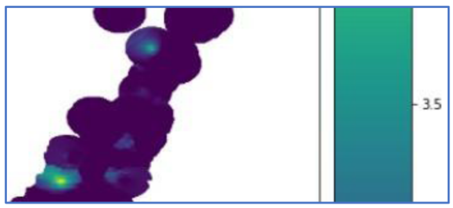


Figure 3—Drifting Tendency Heat Map for Short Pendulum BHA

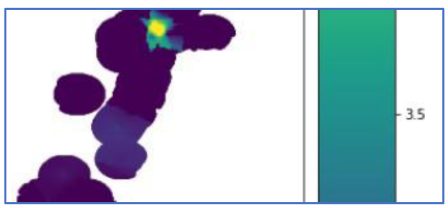


Figure 4—Drifting Tendency Heat Map for Short Packed BHA

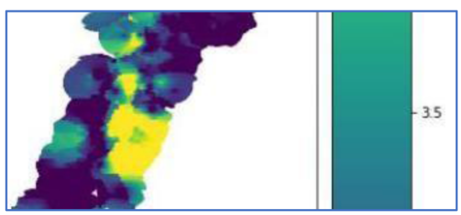


Figure 5—Drifting Tendency Heat Map for Normal Packed BHA BHA

Combining the analysis of the wellbore inclination and Dog-Leg Severity (DLS) based on depth with the BHA utilized for drilling, the software screens out the wells where directional control was available, leaving behind only the wells where uncontrolled angle build-up was observed. As a result, the drift-prone areas are further narrowed down and color-coded to raise a flag. This enables the drilling engineers to select a suitable BHA for drilling.

Discussion of Results

Leveraging advanced data analysis platforms can significantly reduce the time taken to process available information and form meaningful trends. Such platforms have the capability to present the output in visually appealing formats, which make it easy to identify trends. By carefully designing algorithms to focus on the type of analysis required, a lot of time can be saved, freeing up Drilling Engineers to direct their energies to planning mitigation measures for the highlighted risks.

After successfully developing this software application, it was tested by deploying the suggested BHAs in risk-prone areas without compromising drilling performance. In fact, drilling performance improved as maximum drilling parameters could be employed without fear of inclination build-up. In pad wells or in wells where collision issues are a concern, pendulum BHAs have been deployed effectively to maintain verticality; however, deviation survey tools become an essential part of the BHA to monitor inclination. This has led to considerable cost savings by eliminating the need to deploy directional tools for trajectory correction.

It is worth mentioning that once the drifting problem was effectively addressed for drilling deep gas wells, and pendulum BHA was identified as a suitable mitigation measure, the utilization of this assembly as a default has proved to be extremely beneficial. It is observed that if a packed BHA is run, uncontrolled inclination increase is a high risk, and as a result, the use of pendulum BHAs is mandated to avoid any undesirable results.

Conclusions and Recommendations

Utilizing a statistical approach to risk evaluation combined with heatmap visualizations to identify drifting tendency with respect to BHA configurations has proven to be a beneficial method of tackling the drifting problem. Effective algorithms with defined filtering criteria assist in quick data analysis and filtering. The produced heat maps are a quick way of identifying the risks and prompt the deployment of tailor-fit BHAs to solve the issue.

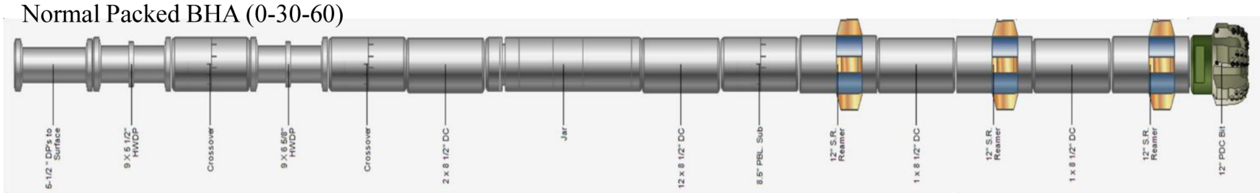
For future developments, it is recommended to expand this approach to cover additional fields where this issue is encountered. Moreover, a more scientific approach can be combined into this methodology to further improve results. This approach would require the acquisition of formation evaluation logs to study the formation characteristics, such as dip angle, heterogeneity, interbedding, etc., to further improve understanding of the underlying geology as well as the interaction of the BHA with the formation.

Acquiring soft skills related to advanced data analysis using available tools is an essential skillset for the future. Utilizing these, as well as Artificial Intelligence (AI), will be a necessity in the future for Drilling Engineers to quickly come up with analyses and recommendations. Therefore, it is highly recommended that engineers consider up-skilling in emerging technologies to maintain a competitive edge both for themselves and the organizations they work for.

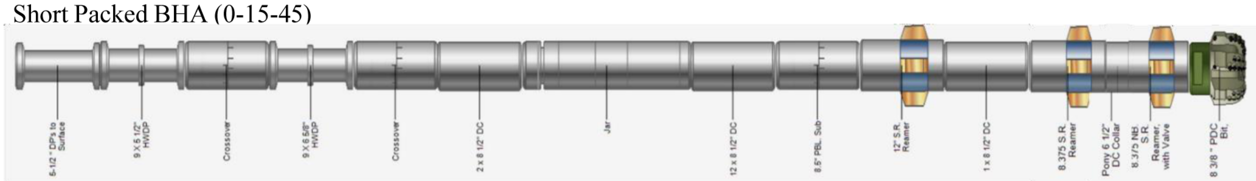
Appendix-1

Various Configurations of Bottom Hole Assemblies (BHAs) used for Drilling Deep Gas Wells

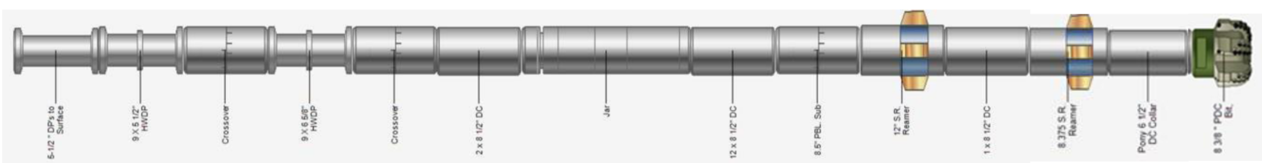
Normal Packed BHA (0-30-60)



Short Packed BHA (0-15-45)



Short Pendulum BHA (15-45)



Since this heat map combines statistical analysis with mathematical computation, it yields an accurate estimate of drilling fluid density that prevents flow while at the same time avoids encountering losses that can seriously hamper progress. In sensitive areas, use of MPD becomes imperative, which has a significant cost impact. To overcome these challenges, the heat map becomes an essential tool that has rescued several wells and is continuously being refined as more data becomes available.

It should be mentioned that the analysis of offset wells with regard to encountered drilling troubles was done manually by the drilling engineering by reviewing the end-of-well reports and the daily drilling reports. This could take several days, and eventually arriving to a decision on the selected mud weight to drill the section would be a subjective call. The automation of this process using the latest available digital technologies is a landmark achievement that reduces this analysis to a few minutes and enables immediate decision making, thereby reducing an engineer's workload and freeing up resources to be applied elsewhere for improving the planning process.